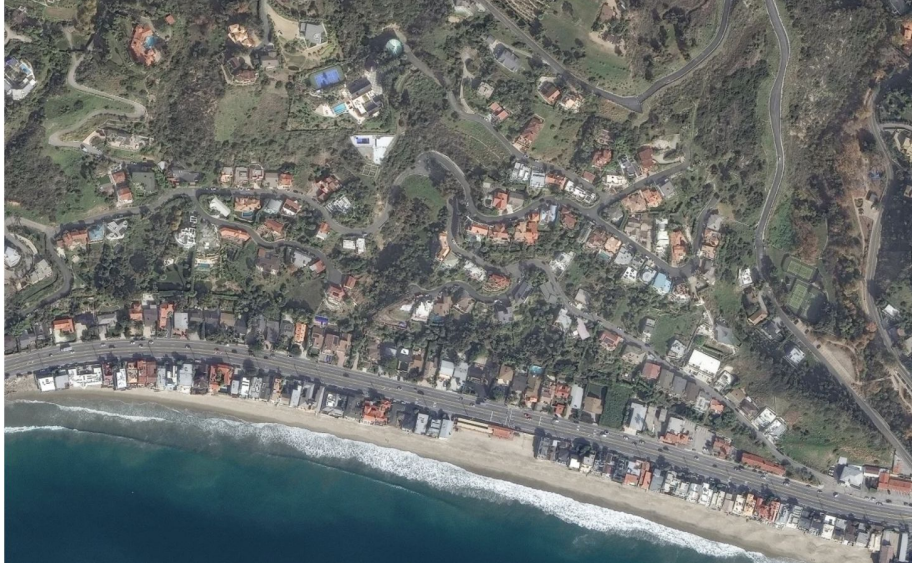


Fire-Flight ML-Driven Aerial Sensing for Wildfire Prediction and Smart Evacuation



Foothill Engineering Club's RC Plane Team

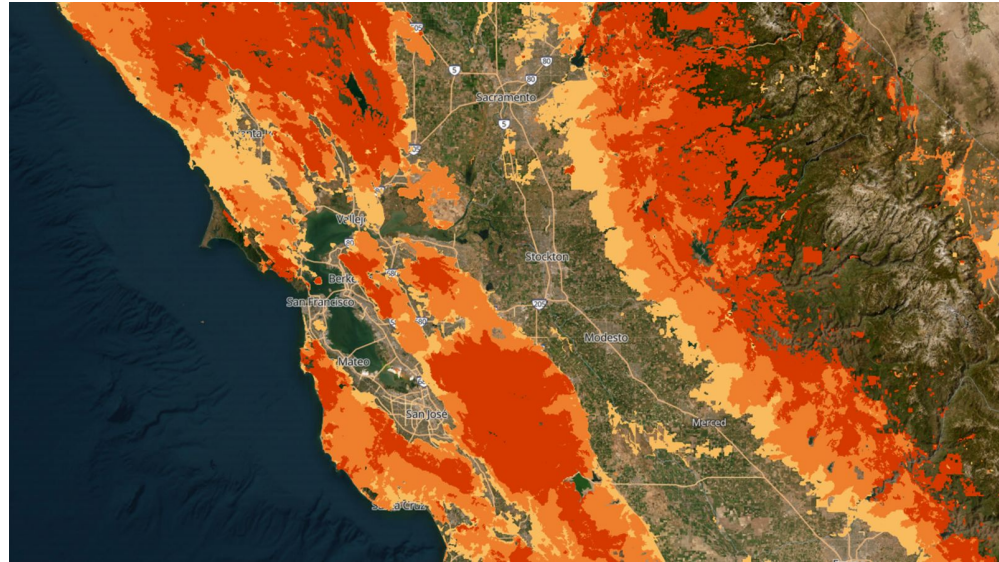
- 1. Wildfire Prediction: Strengths & Gaps**
- 2. Airplane Design for Real-Time Data Collection**
- 3. Translating Diverse Sensor Data into ML Inputs**



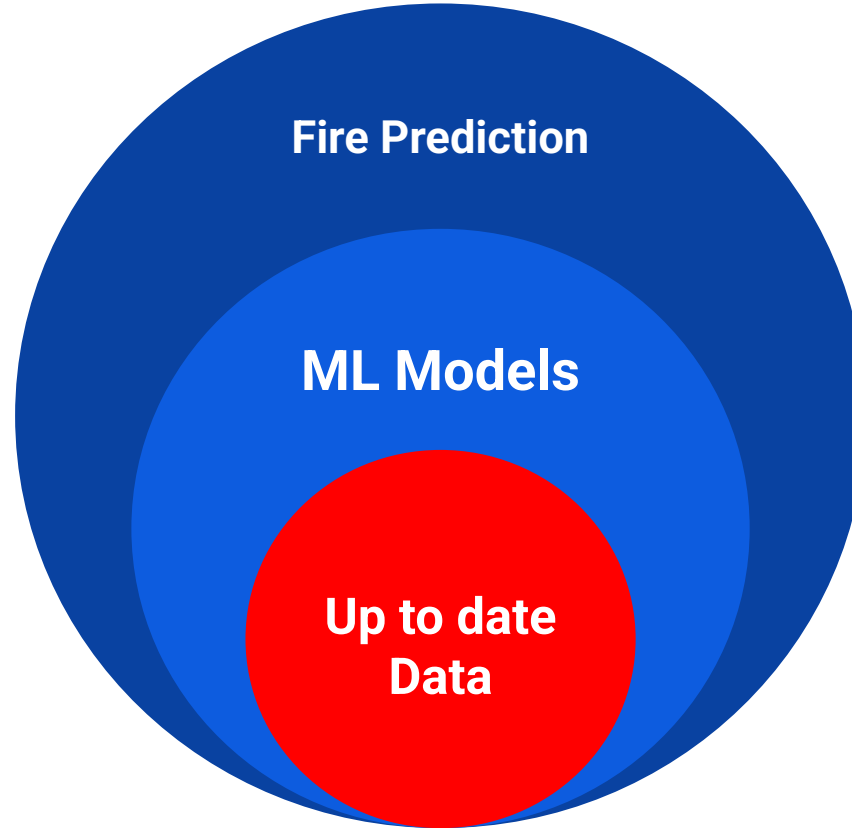
LA Palisade Fires 2025 – Before and After

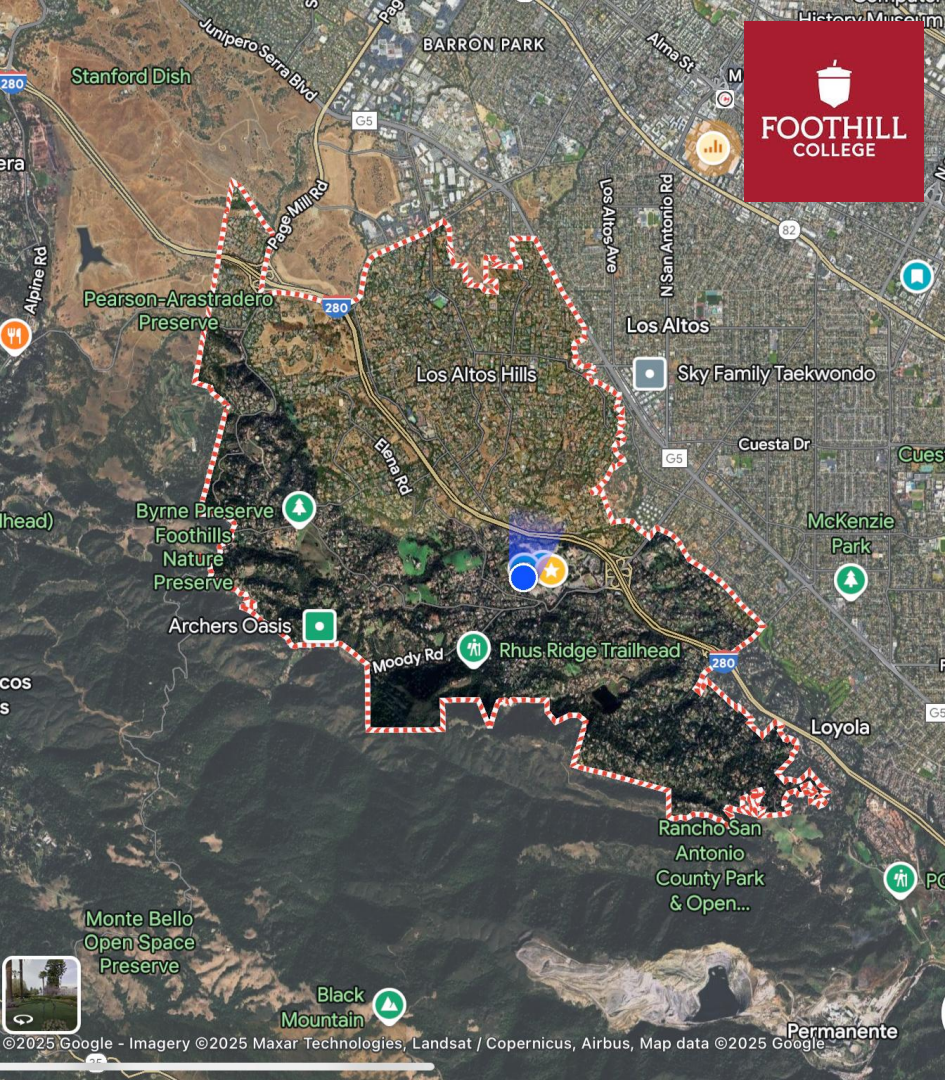


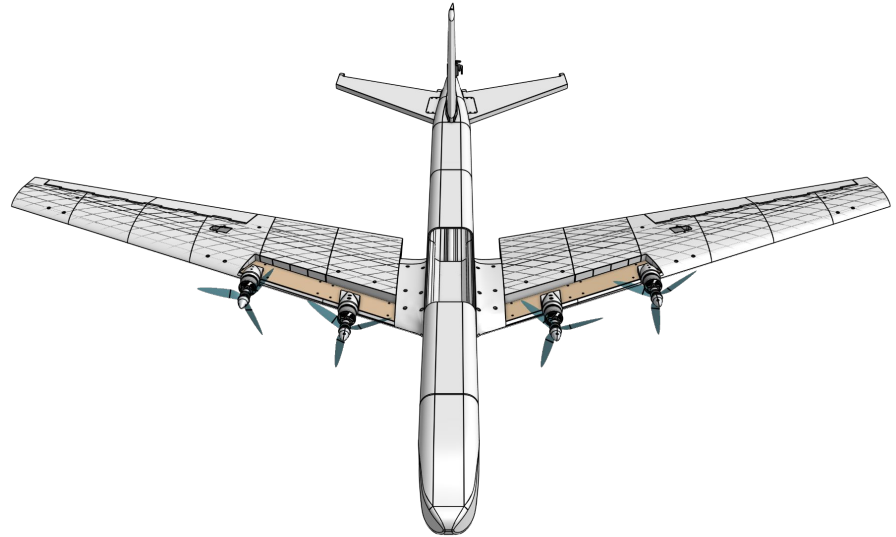
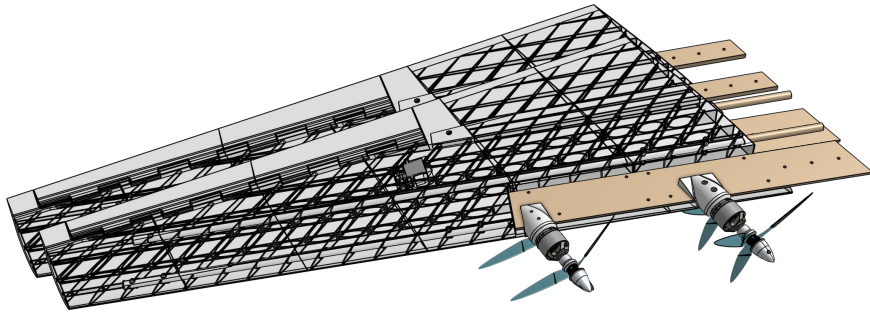
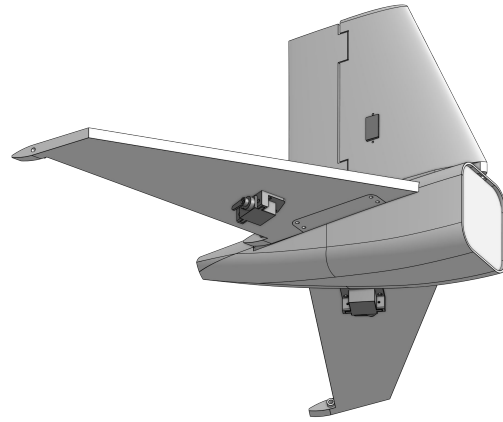
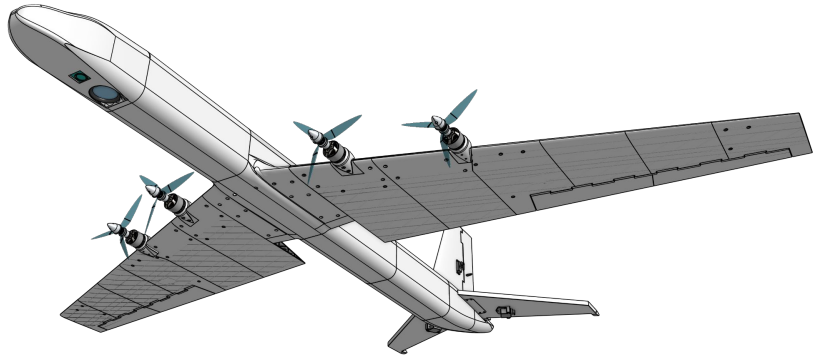
We're here!

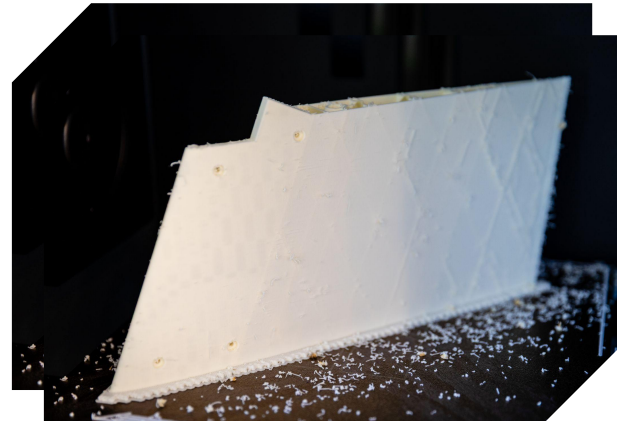
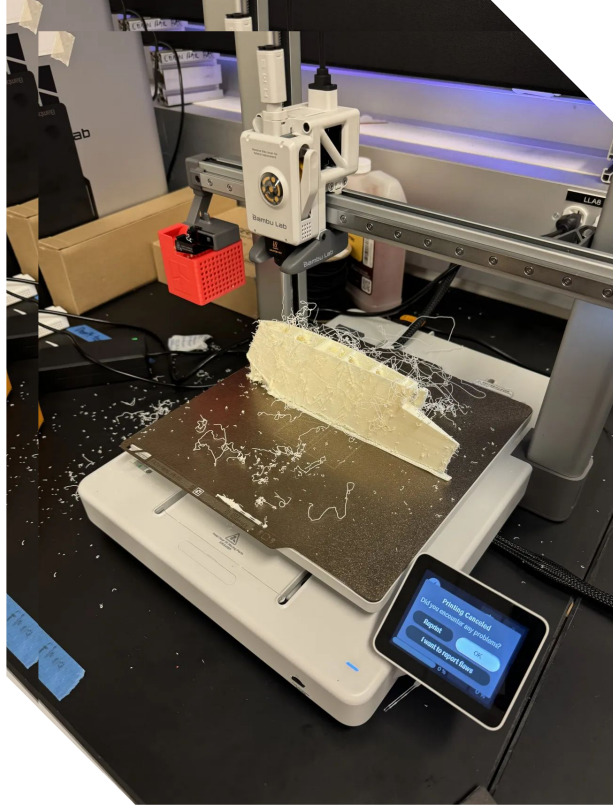


Fire risk map for Central California

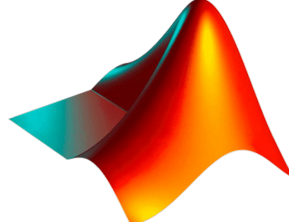


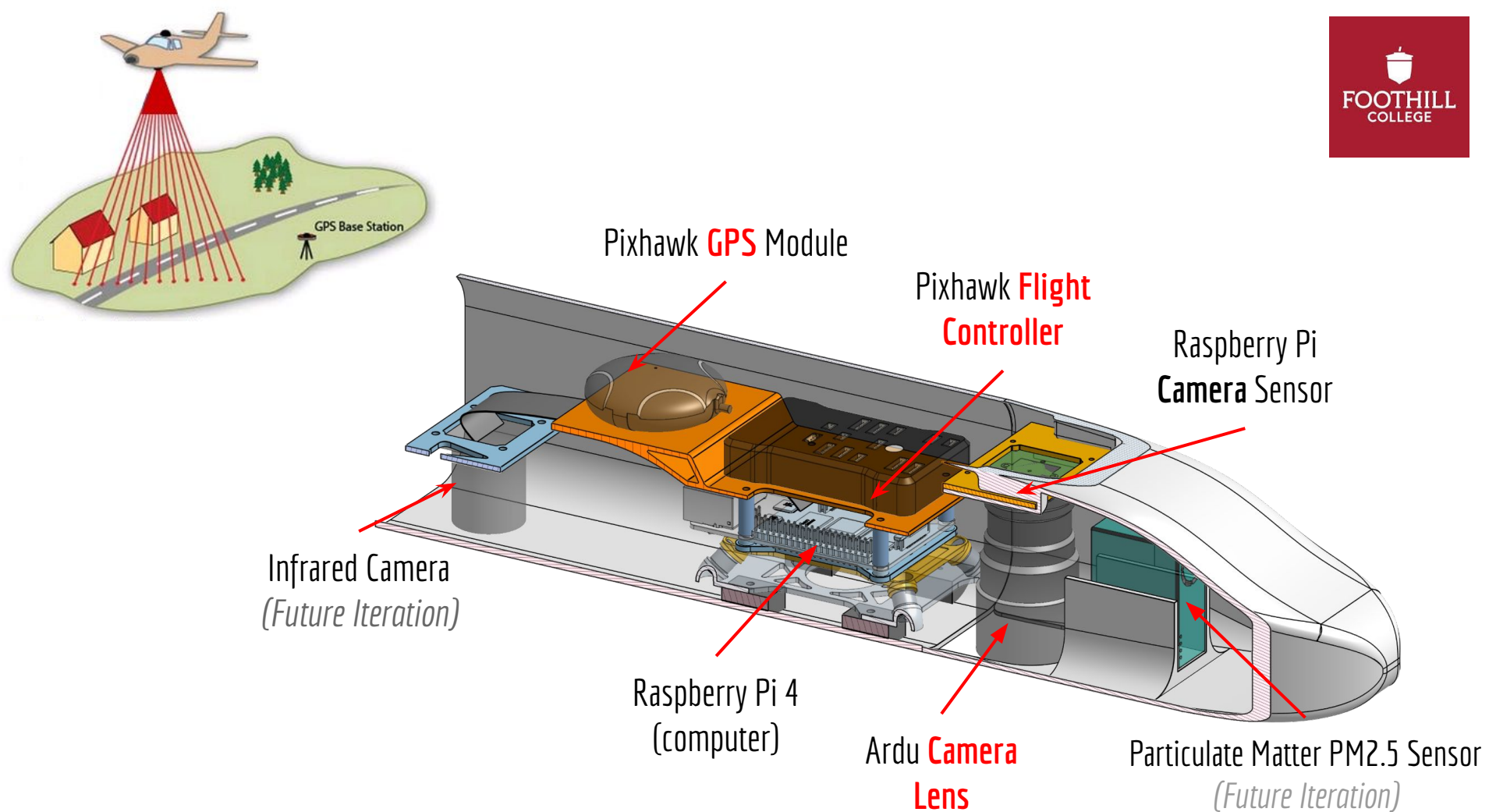


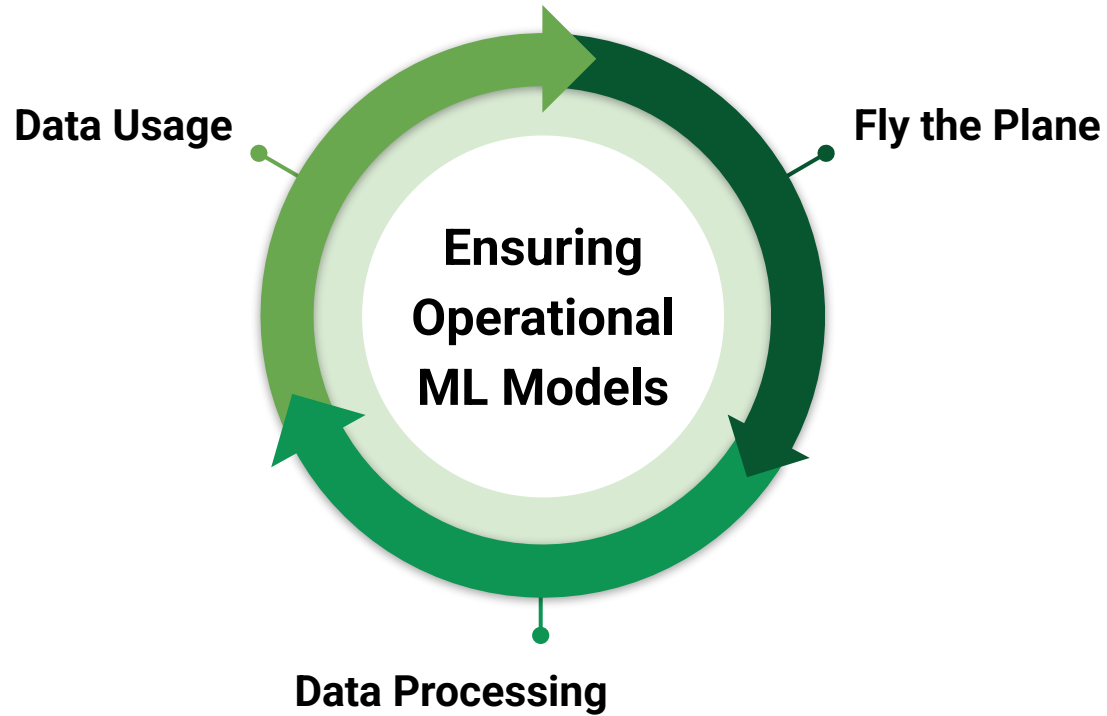




 **SOLIDWORKS**

 **MATLAB®**







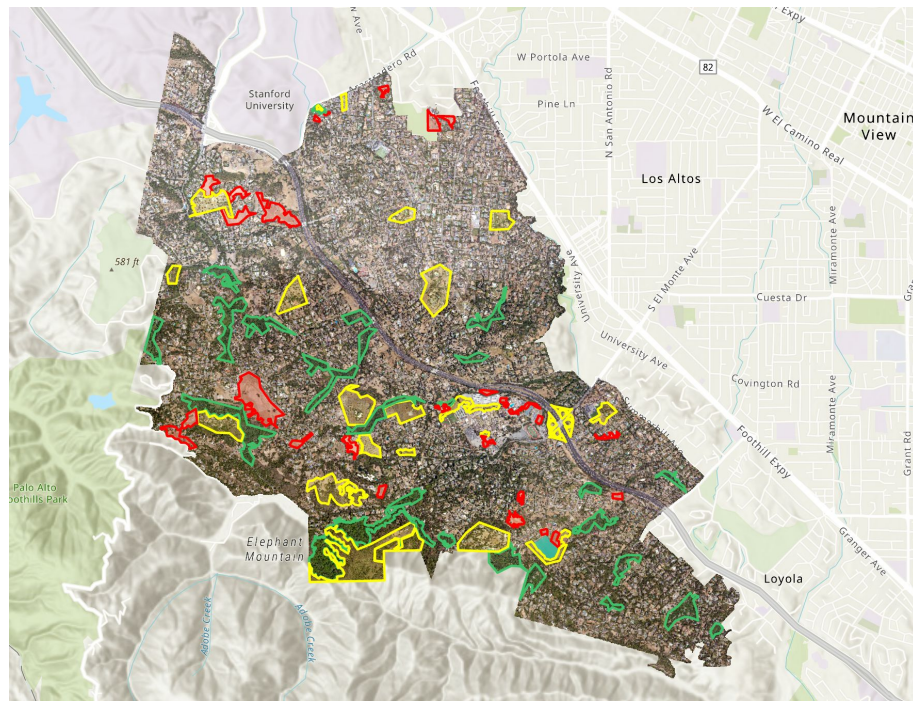
Rothermel Model

$$R = \frac{\text{Heat Source}}{\text{Heat Sink}}$$

**Rate of Fire
Spread**

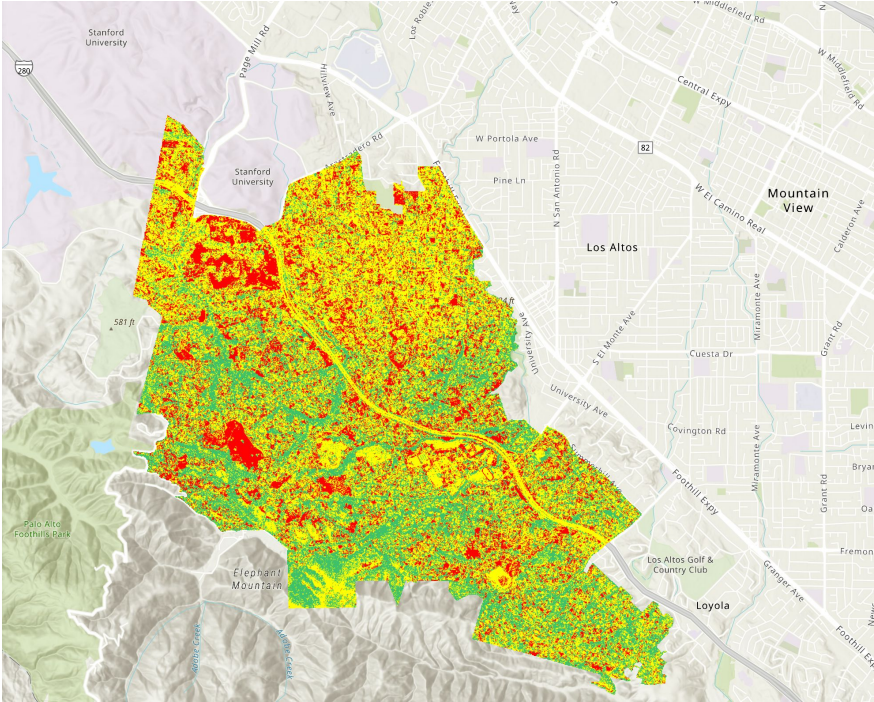


Dead vegetation vs live vegetation



	Risk Level
	Low
	Moderate
	High

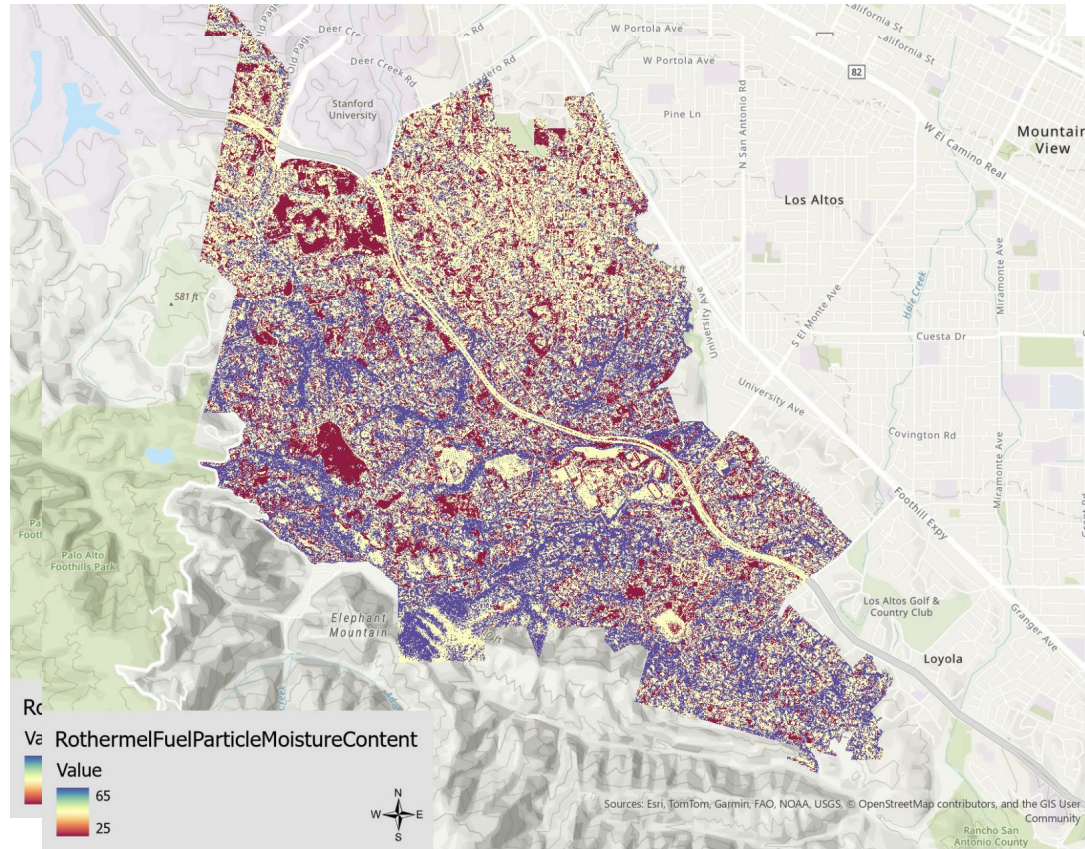
High quality training data for the three risk level categories



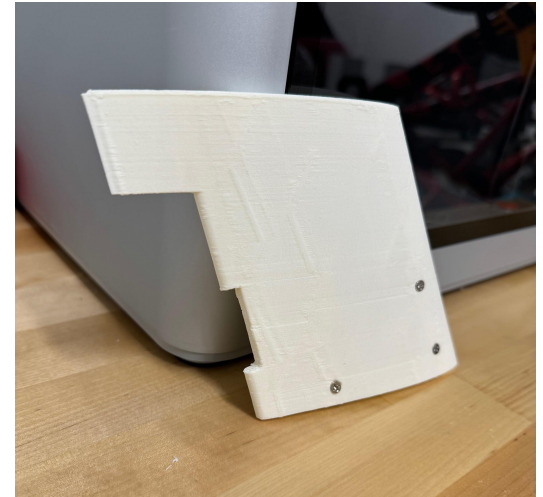
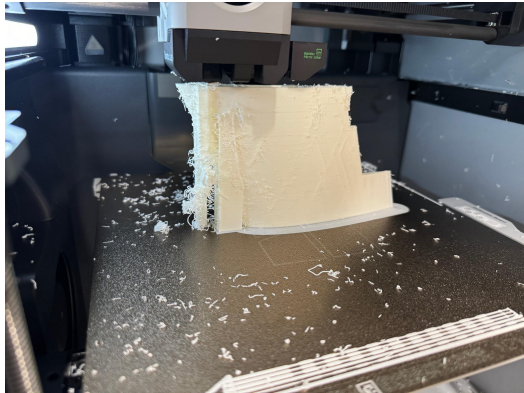
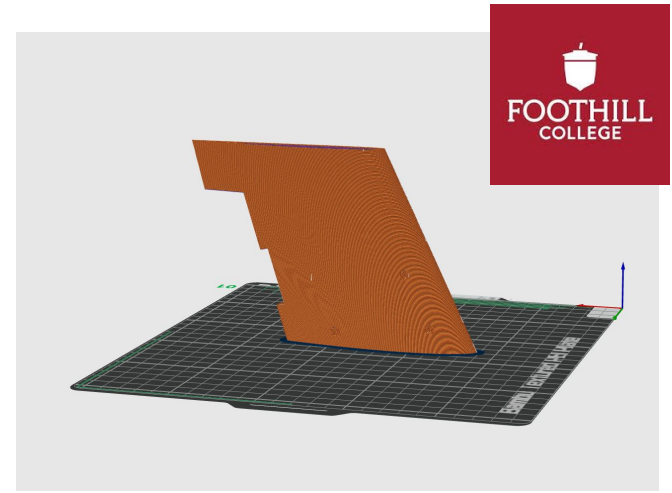
	Risk Level	Moisture Level %
	Low	65
	Moderate	45
	High	25

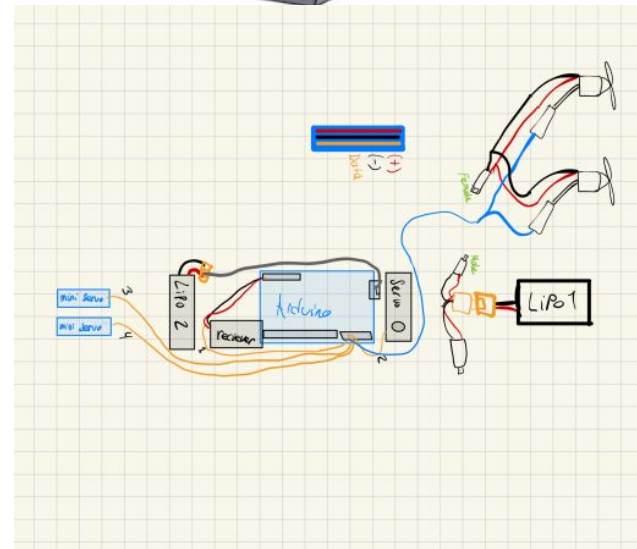
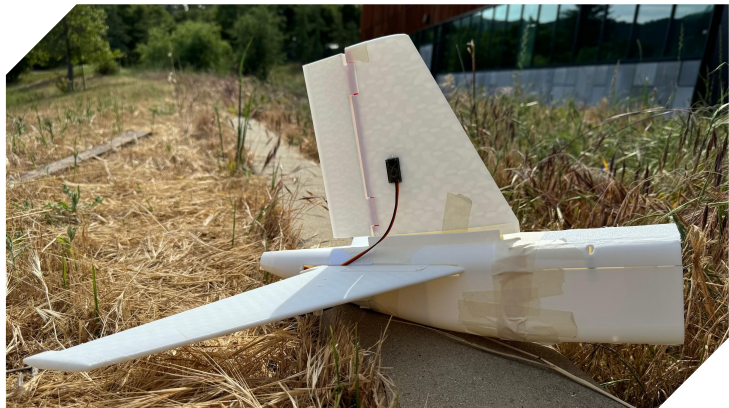
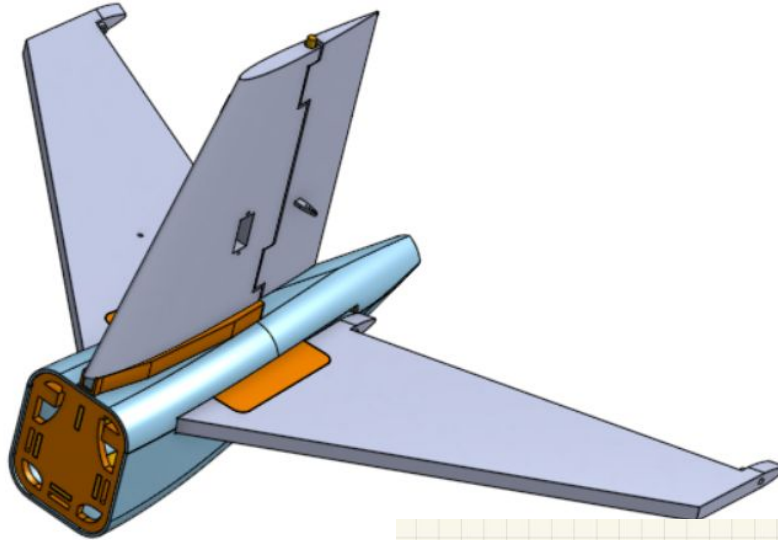
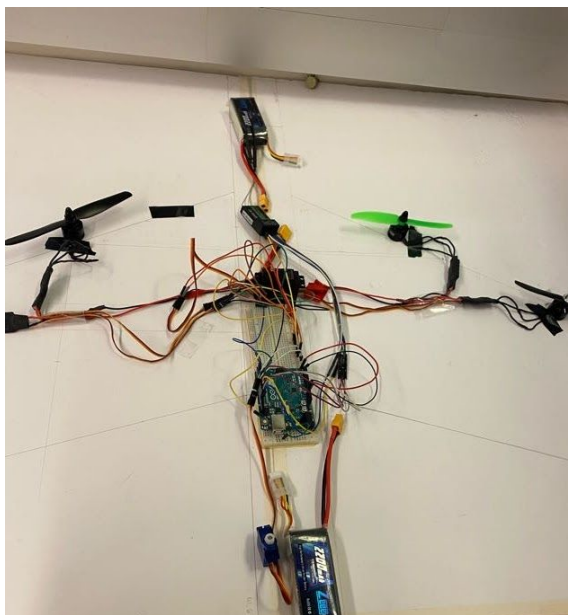
30%

Complete Image Classification for the Three Risk Level Categories



Range of moisture percentages for Los Altos Hills





Thank you!

